MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 19: Applications of the Residue Theorem

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Lecture outline

1. Summation or Real Series
2. The Inverse Laplace Transform

Summation or Real Series. Part 1

Let $P(z)$ be a polynomial of degree at least 2 with real coefficients and no integer zeros (although it may possess other real zeros). Let $z_1, \ldots, z_n$ be the zeros (real or complex) of $P(z)$. Then

$$\sum_{n=-\infty}^{\infty} \frac{1}{P(n)} = -\pi \sum_{j=1}^{m} \operatorname{Res}\left(\frac{\cot(\pi z)}{P(z)}, z_j\right)$$

and

$$\sum_{n=1}^{\infty} (-1)^n \frac{1}{P(n)} = -\pi \sum_{j=1}^{m} \operatorname{Res}\left(\frac{\csc(\pi z)}{P(z)}, z_j\right).$$

We omit the proof and consider some examples.

Example 1. Part 1.

Example: Sum the series

$$\sum_{n=0}^{\infty} \frac{1}{n^2 + a^2},$$

where $a > 0$.

Let $P(z) = a^2 + z^2$. Then $P(z)$ is a polynomial of degree 2, with zeros at $z = \pm ai$. Also, let

$$f(z) = \frac{\cot(\pi z)}{P(z)}$$

Thus

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2 + a^2} = -\pi \left[\operatorname{Res}(f, ai) + \operatorname{Res}(f, -ai)\right]$$

$$\operatorname{Res}(f, ai) = \frac{\cot(\pi ai)}{2ai}$$

and

$$\operatorname{Res}(f, -ai) = \frac{\cot(-\pi ai)}{-2ai} = \frac{\cot(\pi ai)}{2ai}.$$

Example 1. Part 2.

Thus

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+a^2} = -\pi \left[\frac{2\cot(\pi a i)}{2ai} \right] = \frac{\pi i}{a} \cot(\pi a i) =$$

$$= \frac{\pi i}{a} \frac{e^{-\pi a} + e^{\pi a}}{\left(\frac{1}{i}\right)\left(e^{-\pi a} - e^{\pi a}\right)} = \frac{\pi}{a} \left[\frac{e^{-\pi a} + e^{\pi a}}{e^{\pi a} - e^{-\pi a}} \right] = \frac{\pi}{a} \coth \pi a$$

However, this is not the series we want to sum. Noting that $(-n)^2 = n^2$ we have

$$\sum_{n=-\infty}^{\infty} \frac{1}{n^2+a^2} = \sum_{n=-\infty}^{0} \frac{1}{n^2+a^2} - \frac{1}{a^2} + \sum_{n=0}^{\infty} \frac{1}{n^2+a^2} =$$

$$= -\frac{1}{a^2} + 2\sum_{n=0}^{\infty} \frac{1}{n^2+a^2} = \frac{\pi}{a} \coth \pi a$$

Solving for the series that we want we get

$$\sum_{n=0}^{\infty} \frac{1}{n^2+a^2} = \frac{1}{2a^2} + \frac{\pi}{2a} \coth \pi a$$

Example 2

Example

Find

$$\sum_{n=0}^{\infty} \frac{1}{n^2+1}$$

(taking $a = 1$ in the previous example)

$$\sum_{n=0}^{\infty} \frac{1}{n^2+1} = 1 + \frac{1}{5} + \frac{1}{5} + \cdots = \frac{1}{2} + \frac{\pi}{2} \coth \pi$$

Example 2

Example

Show that

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2} = \frac{\pi^2 \cos \pi a}{\sin^2 \pi a}$$

where a is real and different from $0, \pm 1, \pm 2, \ldots$

Let $P(z) = (z+a)^2$ and $f(z) = \dfrac{\csc \pi z}{P(z)}$. Then f has a double pole at $z = -a$ and

$$\operatorname{Res}(f, -a) = \lim_{z \to a} \frac{d}{dz} \left[(z+a)^2 \frac{\csc \pi z}{(z+a)^2} \right] = -\pi \csc \pi a \cot \pi a.$$

Hence

$$\sum_{n=-\infty}^{\infty} \frac{(-1)^n}{(n+a)^2} = \pi^2 \csc \pi a \cot \pi a = \frac{\pi^2 \cos \pi a}{\sin^2 \pi a}.$$

An Inversion Formula for the Laplace Transform.

The **Laplace Transform** of a (real) function is given by

$$L\left[f\left(t\right)\right] = \phi\left(s\right) = \int_0^\infty e^{-st} f\left(t\right) dt$$

for all such s that this integral converges.

The Laplace Transform. Example.

Example

Find the Laplace Transform of $f(t) = e^{at}$

$$L[f(t)] = L[f(t)] = \int_0^\infty e^{-st} e^{at} dt = \int_0^\infty e^{-(s-a)t} dt = \lim_{R\to\infty} \int_0^R e^{-(s-a)t} dt$$

$$= \lim_{R\to\infty} \left. \frac{e^{-(s-a)t}}{a-s} \right|_0^R = \lim_{R\to\infty} \frac{e^{-(s-a)R} - 1}{a-s} = \frac{1}{s-a} \qquad \text{if } s > a$$

. Therefore

$$L\left[e^{at}\right] = \frac{1}{s-a}$$

We may define an inverse L^{-1} operator where $L^{-1}[\phi(s)] = f(t) \iff L[f(t)] = \phi(s)$. Thus

$$L^{-1}\left[\frac{1}{s-a}\right] = e^{at}.$$

The Laplace Transform. Example 2.

Laplace transforms are very useful in solving ordinary and partial differential equations. To illustrate the power of the method, consider the problem:

$$y' + y = e^{2t} \qquad y(0) = -1$$

We take Laplace transforms of both sides of the equation to get

$$L[y'] + L[y] = L\left[e^{2t}\right] = \frac{1}{s-2}$$

Now

$$L[y'] = \int_0^\infty \frac{dy}{dt} e^{-st} dt = ye^{-st}\Big|_0^\infty + \int_0^\infty ye^{-st} dt = sL[y] - y(0)$$

The D.E. then leads to

$$(s+1)L[y] + 1 = \frac{1}{s-2}$$

or

$$L[y] = \frac{-s+3}{(s-2)(s+1)} = \frac{\frac{1}{3}}{s-2} + \frac{-\frac{4}{3}}{s+1}.$$

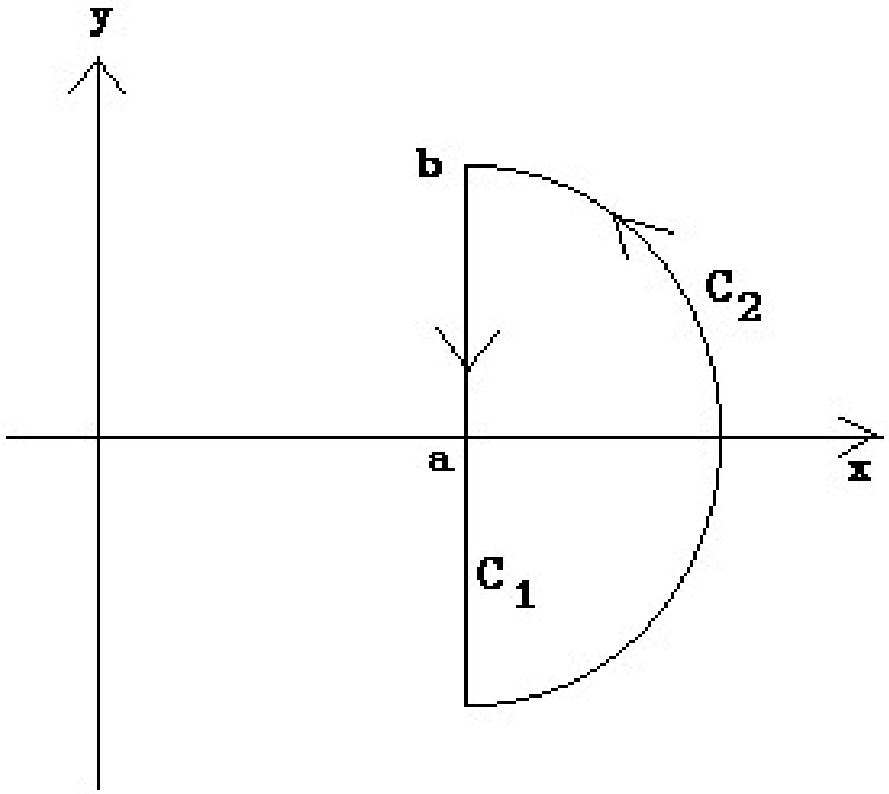
Thus $y(t) = \frac{1}{3}e^{2t} - \frac{4}{3}e^{-t}$.

The Inverse Laplace Transform. Part 1.

This method may be extended to higher order equations and to partial differential equations. Note that in order to get $y(t)$ we had to find $L^{-1}[\phi(s)]$. There exists a general approach to finding $L^{-1}[\phi(s)]$ which is based on the complex variable theory we have developed.

Let $\phi(z)$ be a function of z analytic on the line $x = a$ and in the entire half plane R to the right of this line. Moreover, let $|\phi(z)| \to 0$ uniformly as z becomes infinite through this plane. Let s be any point in the half plane . We can choose a semicircular contour $C = C_1 \cup C_2$ as in the figure on the next slide and apply the Cauchy Integral Formula.

The Inverse Laplace Transform. Part 2.



$$\phi(s) = \frac{1}{2\pi i}\oint_C \frac{\phi(z)}{z-s}dz = \frac{1}{2\pi i}\int_{a+bi}^{a-bi} \frac{\phi(z)}{z-s}dz + \frac{1}{2\pi i}\int_{C_2} \frac{\phi(z)}{z-s}dz$$

The Inverse Laplace Transform. Part 3.

It can be shown that

$$\lim_{b\to\infty}\int_{C_2}\frac{\phi(z)}{z-s}dz=0$$

so that

$$\phi(s)=\frac{1}{2\pi i}\int_{a+\infty i}^{a-\infty i}\frac{\phi(z)}{z-s}dz.$$

Now we shall try to determine the function whose Laplace Transform is $\phi(s)$.

$$L^{-1}[\phi(s)]=f(t)=L^{-1}\left[\frac{1}{2\pi i}\int_{a-\infty i}^{a+\infty i}\frac{\phi(z)}{s-z}dz\right]$$

Hence

$$f(t)=\frac{1}{2\pi i}\int_{a-\infty i}^{a+\infty i}L^{-1}\left[\frac{\phi(z)}{s-z}\right]dz=\frac{1}{2\pi i}\int_{a-i\infty}^{a+i\infty}\phi(z)\,e^{zt}dt$$

assuming that one can interchange the operators above.

The Inverse Laplace Transform. Part 4.

Thus $f(t)$ is given by a line integral in the complex plane taken along a vertical line to the right of all singularities of the transform $\phi(s)$. This complex line integral may be evaluated by the method of residues. A semicircular contour whose diameter is the segment from $a - bi$ to $a + bi$ and whose radius b is large is used.

Theorem

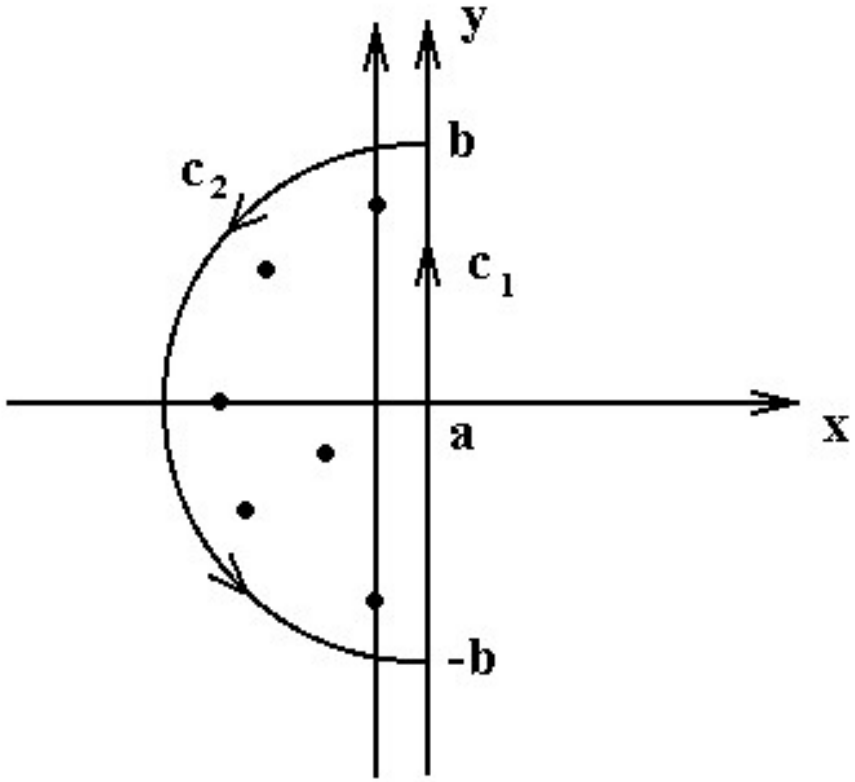
Inverse Laplace Transform *If the Laplace Transform is an analytic function of s except at a finite number of poles each of which lies to the left of the vertical line $Re[s] = a$ and if $s\phi(s)$ is bounded as s becomes infinite through the half plane $Re[s] \leq a$, then*

$$L^{-1}[\phi(s)] = \sum \text{ residues of } e^{st}\phi(s) \text{ at each of its poles.}$$

We omit the proof.

The Inverse Laplace Transform. Part 5.

Remark: Use the contour



The Inverse Laplace Transform. Example

Example

Let a be a positive constant. Find $f(t) = L^{-1}[F]$, where $F(z) = 1/\left(a^2 + z^2\right)$. F has simple poles at $\pm ai$. Now

$$\operatorname{Res}(e^{zt}F(z), ai) = \frac{e^{ait}}{2ai} \qquad \text{and} \qquad \operatorname{Res}(e^{zt}F(z), -ai) = \frac{e^{-ait}}{-2ai}.$$

Hence $f(t) = \dfrac{1}{2ai}\left(e^{ait} - e^{-ait}\right) = \dfrac{1}{a}\sin at.$

Thank you